

Amr Amer

Machine Learning Engineer

M.Sc. Visual Computing, Saarland University

Computer Vision · MLOps · Data Pipelines

Contact

Saarbrücken, Germany

+49 162 6667294

amribrahim.amer@gmail.com

amramer.github.io

linkedin.com/in/amr-amer-2023-cs

github.com/amramer

Skills

Programming

Python, C++ (17/20), SQL, CUDA, Bash, Java

Data Engineering

Apache Airflow, PySpark, pandas, PyArrow, Parquet, DVC (data versioning & lineage), Pandera (data validation), PostgreSQL

MLOps & ML Platform

MLflow (tracking, model registry), Weights & Biases, Optuna, reproducible DVC pipelines, model evaluation & promotion gates

Model Serving & Monitoring

TensorRT, ONNX Runtime, FastAPI, Prometheus, Grafana, edge inference on NVIDIA Jetson (FP16/INT8)

Infrastructure & CI/CD

Docker, Kubernetes, GitHub Actions, AWS (EC2, S3, ECR), SLURM, Enroot, Linux

Machine Learning

PyTorch (DDP), TensorFlow, scikit-learn, Hugging Face Transformers, MONAI, fastai

Computer Vision

OpenCV, YOLO (v8/v10), DETR / RT-DETR, SAM, ByteTrack, DeepSORT,

Profile

Machine Learning Engineer with an M.Sc. in Visual Computing from Saarland University and 4+ years of experience at DFKI. I build production-ready ML systems end-to-end, from versioned, quality-checked data pipelines and distributed GPU training to multimodal generative models, high-performance C++ inference and deployment, with a focus on computer vision.

On the data and MLOps side, I build reproducible pipelines with DVC, Airflow, Spark and MLflow, add automated data-quality gates, and ship models with Docker, Kubernetes and CI/CD on GitHub Actions.

Portfolio: amramer.github.io

Experience

ML Research Engineer — 09/2021 – 12/2025

German Research Center for Artificial Intelligence (DFKI), Saarbrücken

- Owned the data and ML pipelines for multimodal research projects (video, audio, 3D motion, biosignals), from raw recordings to training-ready datasets; versioned data and models with DVC so every result is reproducible from a Git commit.
- Built a camera and LiDAR sensor data pipeline for benchmarking perception models: Airflow orchestration, PySpark processing, and automated schema, synchronization and file-integrity checks that stop bad data before it reaches training.
- Benchmarked and optimized detection models for edge deployment on NVIDIA Jetson, comparing FP32, FP16 and INT8 with TensorRT and ONNX Runtime; integrated models into native C++ inference code and monitored FastAPI services.
- Ran distributed training of transformer and CNN models on multi-node A100 clusters (PyTorch DDP, SLURM, Enroot containers) and used W&B sweeps to cut wasted GPU time.
- Set up MLflow experiment tracking and model registry, PyTest suites and CI/CD with reproducibility checks and promotion gates, so models move to deployment only after passing evaluation.
- Developed a personality-conditioned 3D motion generation model (transformer, VQ-VAE, SMPL-X) with the Max Planck Institute for Informatics, from data preprocessing to cluster training and evaluation.

Tech Stack: PyTorch · C++ · CUDA · TensorRT · ONNX Runtime · DVC · Airflow · PySpark · MLflow · W&B · Docker · Kubernetes · SLURM · GitHub Actions · Prometheus · Grafana

Software Developer (Backend & Data Systems) — 10/2016 – 08/2020

General Authority of Civil Aviation (GACA), Jeddah, Saudi Arabia

- Built secure Java backend services and enterprise data integrations (Oracle SOA Suite) connecting high-volume aviation systems.
- Optimized relational schemas and wrote PL/SQL stored procedures to speed up data throughput for large operational workloads.
- Refactored legacy code into modular, object-oriented components to improve maintainability.

Tech Stack: Java · Oracle SOA Suite · PL/SQL · SQL · SOAP/REST web services · Git

Software Engineering Intern — 04/2016 – 09/2016

Ejada Systems, Alexandria, Egypt

- Developed and tested enterprise J2EE components for backend applications.
- Built and optimized SQL queries and database modules for efficient data access.

Tech Stack: Java · J2EE · SQL · Relational Databases

TrackNet, camera calibration & homography, segmentation, camera–LiDAR fusion, SMPL-X

Model Architectures

CNNs, Vision Transformers (ViT, Swin), Transformers, U-Net, VAE / VQ-VAE, GANs, diffusion models

C++

RAII & smart pointers, move semantics, templates, STL, concurrency (std::thread, std::async), modern C++

Software Engineering

Git, PyTest, REST APIs, performance profiling, Scrum

Languages

- English: C1/C2
- German: B1 (improving)
- Arabic: native

Education

Visual Computing (M.Sc.) — 10/2020 – 04/2024

Saarland University, Saarbrücken, Germany
Master of Science, Grade: 1.9
Thesis Website | Thesis Code | Thesis PDF

- **Master's Thesis:** *Personality-Aware Non-verbal Behavior Generation in Dyadic Interactions*, in collaboration with the Max Planck Institute for Informatics and DFKI.
- Built a multimodal generative model (**transformer, VQ-VAE, SMPL-X/PIXIE**) that generates face, body and hand motion of listener avatars conditioned on personality traits.
- Built the pipeline for data preprocessing, PyTorch training and evaluation on **SLURM clusters** with **NVIDIA A100 GPUs**, including hyperparameter optimization.
- Outperformed a personality-agnostic baseline on UDIVA (**FID 6.15, P-FID 10.31** for face; **FID 43.16, P-FID 87.73** for body and hands); in a user study, **86%** of participants identified the extraverted avatar.

Computer Engineering (B.Eng.) — 09/2011 – 02/2016

Arab Academy for Science and Technology, Egypt
Bachelor of Engineering, GPA: 3.2/4.0

- Thesis: sign language translator using the Kinect sensor, implemented in C++. Demo

Projects

- **AV Sensor DataOps**
  Automated platform that turns raw camera and LiDAR logs into versioned, quality-checked training data, then trains, evaluates, gates and serves a 3D object detector with monitoring.
  (DVC · Airflow · PySpark · Pandera · MLflow · YOLOv8 · ONNX Runtime · FastAPI · Prometheus · Grafana · Kubernetes · GitHub Actions)
- **Badminton-VisionAI**
  End-to-end video analytics pipeline: player and shuttle tracking, shot classification and mini-court projection, with an analytics dashboard and coach report; **92% shot detection accuracy**.
  (OpenCV · YOLOv8 · SAM · TrackNet · Homography · Docker · Streamlit · Plotly)
  Pipeline video Live demo
- **MePheSTO (DFKI research project)**
  ML pipeline over video, speech and biosignal data to derive behavioral markers for psychiatric disorders.
  (PyTorch · Transformers · Multimodal Fusion · Enroot · SLURM · TensorBoard)
- **Semantic Segmentation for Autonomous Driving**
  Segmentation pipeline on BDD100K with optimized preprocessing and training; **mIoU 0.45** (road 0.88, vehicles 0.78) and **30% faster inference**.
  (PyTorch · fastai · Optuna · Weights & Biases)
- **3D Brain Tumor Segmentation**
  3D SegResNet pipeline (MONAI) for MRI segmentation; **mean Dice 0.78** (WT 0.90, TC 0.82) and **20% fewer false positives**.
  (PyTorch · MONAI · 3D CNNs · Sliding-window inference)

More projects and demos: amramer.github.io